

BEYOND STANDALONE PERFORMANCE: WHAT DO TIME-SERIES ANOMALY DETECTORS ADD BEYOND SIMPLE STATISTICS?

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ABSTRACT

Current TSAD benchmarks ask whether a score predicts anomalies, but not what it adds beyond simple statistical signals. We introduce conditional detection utility (CDU), the held-out log-loss reduction from adding a detector score to a statistical reference. We evaluate fixed anomaly-score outputs from nine detectors on 350 univariate series spanning 23 source groups, using leave-one-source-out evaluation. Under the same spline probe, loss, and source weighting, MOMENT-FT and MOMENT-ZS rank fourth and fifth by detector-only utility but seventh and ninth by CDU; TranAD rises from sixth to fourth. POLY and MOMENT-ZS have nearly identical benchmark VUS-PR (0.389 versus 0.383), yet differ by 0.00593 bits of CDU. MOMENT-ZS scores are highly reconstructible from the reference ($R^2 = 0.904$), but rank ninth by CDU; across all nine detectors, reconstructibility and CDU have little rank correspondence (Spearman $\rho = -0.167$). Three probe families preserve the top-two order and top-four membership; five sampling seeds give the same spline ranking, and reference variants preserve the leading group. Standalone performance and beyond-reference contribution are empirically non-interchangeable evaluation axes; CDU makes the latter measurable for heterogeneous TSAD scores.

Index Terms— Time-series anomaly detection, conditional utility, statistical reference, source-held-out evaluation

1. INTRODUCTION

Current TSAD benchmarks evaluate whether an anomaly score predicts anomalies, but not whether it contributes information beyond simple statistical signals already present in the time series. These are different questions. Local variance and deviation can themselves be competitive anomaly scores [1]; a modern detector may therefore score highly while largely reproducing them, whereas a lower-ranked detector may provide complementary anomaly evidence. Benchmarks such as TSB-AD [2] expose standalone performance but not this added contribution.

This motivates two distinct questions. First, how well can a statistical reference B reconstruct a detector score S_D ? Second, given B , how much additional label-prediction utility

does S_D provide? The first measures statistical overlap; the second measures whether the detector contributes anomaly evidence beyond that overlap. Difference alone is insufficient: independent noise can differ from the reference without helping.

We introduce conditional detection utility (CDU) as an evaluation axis missing from current TSAD benchmarks. CDU compares held-out log-loss with and without the detector score while retaining the same statistical reference. It complements volume under the precision-recall surface (VUS-PR) [3]: standalone evaluation asks whether a score predicts anomalies; conditional evaluation asks what it adds beyond a reference. POLY and MOMENT-ZS have nearly identical VUS-PR (0.389 versus 0.383), yet under our primary spline evaluation POLY provides 0.00593 bits more CDU. The distinction also persists when detector-only utility and CDU are compared under the same log-loss. MOMENT-ZS is highly reconstructible from the reference ($R^2 = 0.904$), demonstrating that statistical overlap and added label utility are different notions.

Our contributions are threefold. First, we identify a blind spot in standard TSAD evaluation: standalone performance cannot distinguish statistical redundancy from complementary label signal. Second, we introduce CDU, a paired source-held-out measure of the predictive value added by a fixed detector score beyond a statistical reference; its unrestricted population counterpart is conditional mutual information. Third, across 350 series and nine detectors, we show that conditioning changes detector comparisons, statistical reconstructibility does not determine added label utility, and the resulting leading structure persists across probes, seeds, and reference variants.

2. RELATED WORK

Predictive $\mathcal{V}$ -information describes information accessible to a chosen predictor family [4]. Conditional probing measures information beyond a baseline representation [5], while predictive variable importance compares risks with and without features [6]. We instantiate this principle for heterogeneous TSAD scores through a common score interface, source-held-out fitting, matched standalone/conditional losses, and source-balanced aggregation.

In TSAD, comparisons with one-line statistical scorers reveal overlap in benchmark performance [1]. Work on time-series foundation models also examines whether forecast-error features add value beyond window statistics under drift [7]. Our focus is a common, score-level conditional evaluation across detector families, with matched losses and source-held-out testing.

3. CONDITIONAL DETECTION UTILITY

3.1. Paired predictive evaluation

Let $Y \in \{0, 1\}$ denote an anomaly label, $B \in \mathbb{R}^p$ a statistical reference, and S_D a detector score. For each held-out source, fit two probability predictors from the same probe family $\mathcal{Q}$: one receives B , and the other (B, S_D) . Both use the same training points, weights, and fitting settings. The operational estimate is

$$\widehat{\text{CDU}}_{\mathcal{Q}}(D | B) = \widehat{L}_B - \widehat{L}_{BD}, \quad (1)$$

where $\widehat{L}_B$ and $\widehat{L}_{BD}$ are their paired held-out log-losses in bits. Positive CDU means that the detector improves label prediction given the reference. Figure 1 summarizes the comparison.

For a matched standalone comparison, define $U_D = \widehat{L}_0 - \widehat{L}_D$. Here the detector-only probe receives S_D , and the null predicts weighted anomaly prevalence estimated only from training sources. Thus U_D and CDU use the same loss but different reference information.

3.2. Population interpretation

Define $\mathcal{R}^*(X) = \inf_q \mathbb{E}[-\log_2 q(Y | X)]$, where q ranges over conditional probability distributions.

Proposition 1. Under a common joint distribution and unrestricted predictors,

$$\begin{aligned} \text{CDU}^*(D | B) &= \mathcal{R}^*(B) - \mathcal{R}^*(B, S_D) \\ &= I(Y; S_D | B). \end{aligned} \quad (2)$$

Proof. The cross-entropy decomposition gives

$$\begin{aligned} \mathbb{E}[-\log_2 q(Y | X)] &= H(Y | X) \\ &\quad + \mathbb{E}_X[D_{\text{KL}}(P_{Y|X} \| q_{Y|X})]. \end{aligned} \quad (3)$$

The KL term is nonnegative and vanishes for $q = P$, giving $\mathcal{R}^*(X) = H(Y | X)$. Taking the difference proves the identity. $\square$

Consequently, population CDU is nonnegative and is zero whenever $S_D = g(B)$: a score determined by the reference adds no information. Finite-probe CDU is an operational, predictor-family-relative estimate of added predictive utility; fitted estimates may be negative under finite-sample transfer.

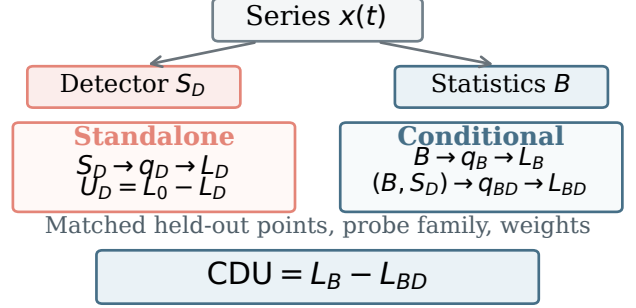


Fig. 1. One series yields a statistical reference and detector score. Standalone utility uses the score alone; CDU compares paired probes with and without it.

3.3. Common score representation

Each curve is aligned to the same time indices as its labels. We evaluate within-series anomaly ordering through average ranks, removing detector-specific score spacing. For a length- T_i curve, ranks start at one:

$$\tilde{s}_{it} = \frac{\text{rank}_{\text{avg}}(s_{it}) - 0.5}{T_i}. \quad (4)$$

The statistical reference uses the same rank representation. Ties are preserved, and constant curves map to 0.5. Rank normalization is part of the evaluation definition: it measures the label information added by a detector's anomaly ordering beyond the rank-normalized statistical reference. The transform is label-free; B and S_D below denote these probe inputs.

3.4. Leave-one-source-out probing

A source is a benchmark dataset group. Each outer fold excludes all series of one source from probe training. Labels from the remaining sources supervise the probes; held-out labels are used only to compute losses. Detector outputs and the statistical reference remain fixed. This strict split prevents probes from exploiting sibling series from the held-out source.

Let G_{tr} be the number of training sources and n_g the number of series in training source g . If m_i points are retained from series i and $N_s = \sum_i m_i$, assign each retained point weight

$$w_{it} = \frac{N_s}{G_{\text{tr}} n_g m_i}. \quad (5)$$

This gives equal total weight to sources and equal weight to series within a source. Full-data training sets $m_i = T_i$; sampled training uses $m_i = \min(T_i, 2048)$. Weights sum to N_s . A common basis-only predictor supplies the reference loss for all detectors within each fold.

3.5. Aggregation and uncertainty

For input setting $M \in \{0, B, D, BD\}$, let p_{it}^M be the held-out anomaly probability. With binary log-loss $\ell(y, p) = -y \log_2 p - (1 - y) \log_2 (1 - p)$, compute

$$L_i^M = \frac{1}{T_i} \sum_{t=1}^{T_i} \ell(y_{it}, p_{it}^M). \quad (6)$$

All test points are retained. Average first within each source and then equally over G_{all} sources:

$$L_g^M = \frac{1}{n_g} \sum_{i \in g} L_i^M, \quad \hat{L}_M = \frac{1}{G_{\text{all}}} \sum_g L_g^M. \quad (7)$$

Our study has $G_{\text{all}} = 23$ and $G_{\text{tr}} = 22$. Paired source-bootstrap 95% intervals quantify held-out source variation conditional on the fitted outer-fold probes.

4. EXPERIMENTAL SETUP

Data and detectors. We evaluate 350 univariate series from TSB-AD-U [2], following the evaluation population in [1]. Source groups follow the dataset identifier in each filename; MSL and SMAP are distinct, and UCR is grouped as one archive. We use SubPCA and POLY benchmark implementations [2], fine-tuned (FT) and zero-shot (ZS) MOMENT [8], M2N2 [9], TranAD [10], TimesNet [11], FITS [12], and AnomalyTransformer [13]. All 350 curves are retained for every detector.

Statistical reference. The 31 fixed, label-free features span local variance, range, prediction deviation, absolute difference, robust dispersion, and spectral entropy. Variance and mean-deviation scorers are motivated by [1]; the remaining families broaden the reference without selecting features on held-out labels.

Probes and comparisons. We use additive cubic-spline logistic regression as the primary probe, with linear logistic and histogram gradient boosting (HGB) as robustness probes. All three use the same uniformly sampled, label-independent training points (at most 2048 per series), source/series weights, and full test curves. Logistic probes use fixed inverse L2 strength $C = 0.1$; spline has four fixed knots on $[0, 1]$, and HGB is a shallow, 32-iteration tree ensemble. No extra class reweighting or hyperparameter search is applied. VUS-PR and CDU use the same detector curves.

Reconstruction and robustness. For score reconstruction, source-held-out ridge regression predicts each detector’s ranked score from the ranked 31-feature reference ($\alpha = 1$); training uses the same 2048-point cap and held-out curves are evaluated in full. We report source-macro R^2 and Spearman correlation. Robustness repeats the spline comparison over five training-sampling seeds, seven leave-one-family-out and cumulative 7–29-feature references, and alternative detector-score interfaces while keeping the ranked reference fixed.

These interfaces use within-series min–max scaling or the Gaussian CDF of within-series z-scores.

Controls. Controls include a duplicated reference feature, independent standard-normal noise, and a rank-transformed synthetic score $Z + 2Y$, with independent $Z \sim \mathcal{N}(0, 1)$, representing redundancy, irrelevance, and added label signal, respectively. Only the synthetic control uses labels.

5. RESULTS AND INTERPRETATION

5.1. Standalone performance and added utility tell different stories

Table 1 and Fig. 2a compare detector-only utility U_D with CDU under the same spline probe, log-loss, held-out sources, and weights. MOMENT-FT and MOMENT-ZS fall from U_D ranks 4 and 5 to CDU ranks 7 and 9, while TranAD rises from 6 to 4. Across linear, spline, and HGB probes, the U_D /CDU rank correlations are 0.967, 0.667, and 0.500. The distinction also appears alongside benchmark VUS-PR: M2N2 and TranAD move from VUS ranks 5 and 6 to CDU ranks 2 and 4.

POLY and both MOMENT variants have similar VUS-PR (0.3832–0.3893), yet POLY exceeds MOMENT-FT by 0.00505 bits of spline CDU [0.00172, 0.00860] and MOMENT-ZS by 0.00593 [0.00241, 0.00961]. Both differences remain positive after a joint max-standardized bootstrap correction over three pairwise POLY/MOMENT comparisons across the three probes.

5.2. Score reconstructibility does not determine added label utility

Source-held-out ridge regression reconstructs MOMENT-FT and MOMENT-ZS scores strongly (source-macro $R^2 = 0.799$ and 0.904 ; Spearman 0.895 and 0.951), but M2N2 and TranAD weakly ($R^2 = 0.054$ and -0.002). Across nine detectors, reconstruction R^2 and spline CDU have Spearman correlation -0.167 (Fig. 2b). Statistical overlap and added label utility show little alignment in this detector set: a hard-to-reconstruct score helps only if its residual information improves label prediction.

5.3. The conditional comparison is reproducible

SubPCA and M2N2 remain first and second across linear, spline, and HGB; TranAD and POLY complete the same top-four set (Fig. 2c). Pairwise rank correlations are 0.717–0.867. Spline and HGB extract substantially more predictive signal from the statistical reference than linear logistic: their basis losses are 0.2891 and 0.2869 bits versus 0.3375. Expanding linear training from sampled to full points barely changes basis loss (0.337462 versus 0.337447), indicating that the gap is driven by probe expressivity rather than additional training points.

Table 1. Primary spline results for nine fixed detector scores. VUS-PR is series-macro; U_D and CDU are in bits. Parentheses give ranks; brackets give source-bootstrap 95% intervals.

Detector	VUS-PR (rank)	U_D (rank)	CDU [95% CI]	CDU rank
SubPCA	0.4224 (1)	0.04060 (1)	0.01509 [0.00256, 0.02964]	1
POLY	0.3893 (2)	0.02700 (3)	0.00604 [0.00246, 0.00969]	3
MOMENT-FT	0.3864 (3)	0.01907 (4)	0.00099 [-0.00001, 0.00208]	7
MOMENT-ZS	0.3832 (4)	0.01832 (5)	0.00011 [-0.00088, 0.00113]	9
M2N2	0.2870 (5)	0.03090 (2)	0.01272 [0.00075, 0.02614]	2
TranAD	0.2754 (6)	0.01438 (6)	0.00484 [-0.00162, 0.01263]	4
TimesNet	0.2627 (7)	0.00591 (8)	0.00133 [-0.00015, 0.00300]	5
FITS	0.2464 (8)	0.00598 (7)	0.00109 [-0.00008, 0.00221]	6
AnomalyTransformer	0.1118 (9)	-0.00005 (9)	0.00015 [-0.00129, 0.00167]	8

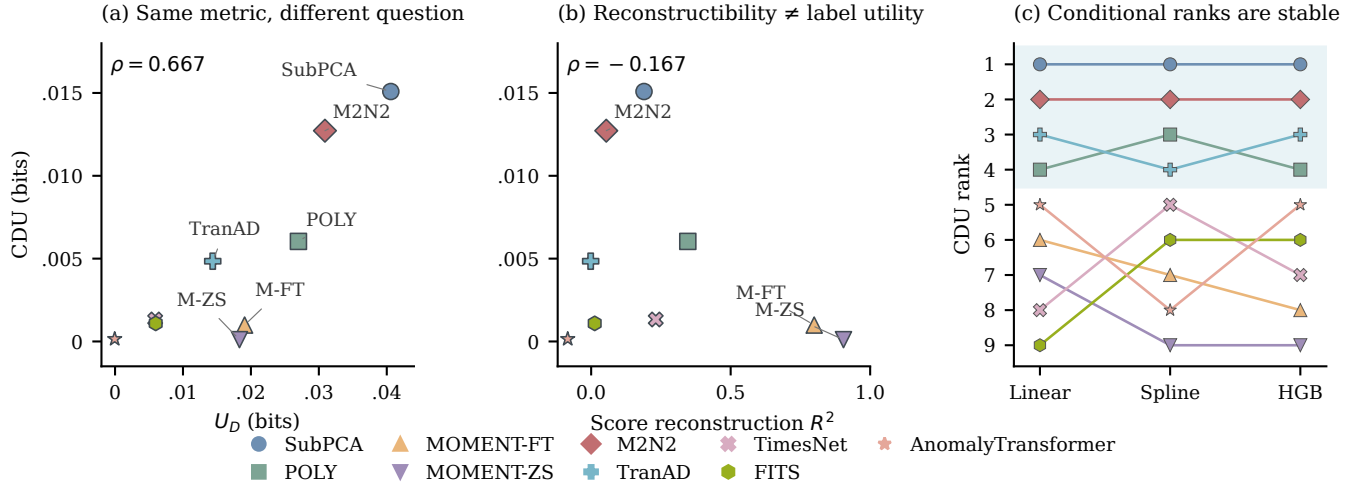


Fig. 2. The benchmark blind spot and its consequences. (a) Detector-only and conditional utility use the same spline probe and log-loss yet differ; (b) statistical reconstructibility does not determine added label utility; (c) the conditional ordering is stable across probes. The top-two order and top-four membership (shaded) persist.

Five training-sampling seeds give identical spline ranks. Seven reference-family removals yield rank correlations of 0.917–1.000; cumulative references with 7–29 features retain the full-reference top four ($\rho = 0.933$ –1.000). A Gaussian-CDF-transformed z-score interface also retains that group ($\rho = 0.933$). Min-max scaling preserves detector-specific score spacing and changes the broader ranking ($\rho = 0.517$), moving POLY from rank 3 to 9. This contrast motivates the common rank interface: our primary CDU comparison concerns anomaly ordering rather than detector-specific score geometry.

5.4. Controls validate the intended behavior

Matched duplicate and independent-noise controls remain near zero, with no negative-control interval entirely above zero. Every label-informed synthetic control is clearly positive; spline synthetic CDU is 0.09220 bits [0.06819, 0.11712]. CDU separates redundant and irrelevant scores from comple-

mentary label signal.

6. DISCUSSION AND CONCLUSION

TSAD benchmarks measure anomaly alignment; CDU measures added utility beyond a statistical reference. Across nine detectors, matched-loss comparisons change ranks, and score reconstruction separates statistical overlap from added label utility. The leading structure persists across probes, seeds, and tested references.

CDU is intentionally reference- and probe-relative and does not assign architectural causes. Reporting it alongside standalone performance separates two questions that current benchmark rankings conflate: whether a detector works, and what it adds beyond simple statistical structure.

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